Age Estimation from Speech: A Data Science Approach

Elias Noorzad, s314515
Parastoo Alavi, s789012
Politecnico di Torino

***Abstract*—This is the abstract of your paper. It summarizes the problem, methodology, and key findings. Keep it concise, typically within 150–250 words.**

I. Problem Overview

This project aims to develop a system for predicting the age of a speaker based on features extracted from their speech. The data set consists of 3,624 samples, divided into a development set with 2,933 samples and an evaluation set with 691 samples. Each sample corresponds to a spoken sentence, with the speaker's age provided as the target variable, represented as a continuous numerical value.

The data set includes the following features:

Acoustic Features

- Mean pitch, maximum pitch, and minimum pitch (measured in Hz): Capture variations in the speaker's vocal frequency.
- Jitter: Measures pitch instability.
- Shimmer: Captures amplitude instability.
- Energy: Represents the overall power of the speech signal.
- Zero-crossing rate (ZCR) mean: Indicates the rate at which the signal changes sign, reflecting tonal changes.
- Spectral centroid mean: Represents the "center of mass" of the frequency spectrum.
- Harmonic-to-noise ratio (HNR): Measures the proportion of harmonic sound to noise, indicating voice quality.
- Duration of silence: Refers to the total time in seconds where no speech is detected.

Linguistic Features

- Number of words: Provides insights into the amount of content spoken.
- Number of characters: Reflects the length of the spoken content.
- Number of pauses: Indicates the speaker's pause patterns and fluency.
- Tempo: Measured in beats per minute (BPM), it reflects the speaker's speaking rate.

Demographic Metadata

- Gender: Captures the speaker's gender information.
- Ethnicity: Captures the speaker's ethnic background to study potential correlations with speech features and age.

These features were extracted to comprehensively capture the acoustic and linguistic properties of the speech signal, facilitating age prediction. Studies have shown that acoustic characteristics such as pitch, jitter, HNR and energy are strong indicators of the age of a speaker due to physiological changes in the vocal cords over time [1].

In addition to acoustic features, linguistic characteristics such as speech rate (tempo) and demographic metadata, including ethnicity, have been shown to significantly contribute to age estimation accuracy. These features capture not only physiological aspects but also cultural and linguistic variability in speech patterns, which play a critical role in predicting the age of the speaker [2].

Similarly, frequent pauses in speech are significant indicators of speaker age, reflecting age-related declines in respiratory and vocal function. Older adults tend to pause more often as a compensatory mechanism, leading to shorter utterances and disrupted linguistic flow. Additionally, listeners often associate frequent pauses with characteristics like frailty or forgetfulness, making pauses both physiologically relevant and socially significant for age estimation [3].

II. Proposed Approach

A. Preprocessing

The preprocessing step ensures that the data is prepared for modeling by addressing inconsistencies, scaling numerical features, and encoding categorical variables. The following steps were applied:

1) **Feature Selection**: Initially, the dataset consisted of 20 columns. Features such as *id*, *path*, and *sampling_rate* were removed because they showed no meaningful correlation with the target variable, *age*. These features do not contribute to the prediction task, leaving a total of 17 columns that are relevant for modeling.
2) **Data Cleaning**: A typo in the *gender* feature, where "female" was misspelled as "famale" in the evaluation set, was corrected to ensure consistency. Additionally, certain features such as *tempo* were not of the correct data type (e.g., float) and were converted to the appropriate type before normalization.
3) **Data Splitting**: The development set was split into a *training set* (80%) and a *test set* (20%) using the train-

test-split method to evaluate the model's performance during training.

4) **Normalization**: Numerical features were normalized using the *z-score normalization* technique to standardize their distribution. Specifically, the following features were normalized: *mean_pitch*, *max_pitch*, *min_pitch*, *jitter*, *shimmer*, *energy*, *zcr_mean*, *spectral_centroid_mean*, *tempo*, *hnr*, *num_words*, *num_characters*, *num_pauses*, and *silence_duration*. The *fit_transform* method was applied to the training set, while the *transform* method was used for the test and evaluation sets to avoid data leakage.
5) **Categorical Encoding**: The categorical features *gender* and *ethnicity* were encoded using *one-hot encoding*. After encoding, the total number of columns increased to 167.

These preprocessing steps ensured that the data was clean, consistent, and prepared for effective model training and evaluation.

III. Results

Present your findings, include tables and graphs.

IV. Discussion

Interpret your results and discuss their implications.

V. Conclusion

Summarize the work and suggest directions for future research.

References

[1] C. T. Ferrand, "Harmonics-to-noise ratio: An index of vocal aging," *Journal of Voice*, vol. 16, no. 4, pp. 480–487, 2002. [Online]. Available: https://doi.org/10.1016/S0892-1997(02)00123-6

[2] D. Jiao, V. Watson, S. G.-J. Wong, K. Gnevsheva, and J. S. Nixon, "Age estimation in foreign-accented speech by non-native speakers of English," *Speech Communication*, vol. 106, pp. 118–126, 2019. [Online]. Available: https://doi.org/10.1016/j.specom.2018.12.005

[3] J. E. Huber, "Effects of utterance length and vocal loudness on speech breathing in older adults," *Respiratory Physiology and Neurobiology*, vol. 164, no. 3, pp. 323–330, Dec. 2008. [Online]. Available: https://doi.org/10.1016/j.resp.2008.08.007.